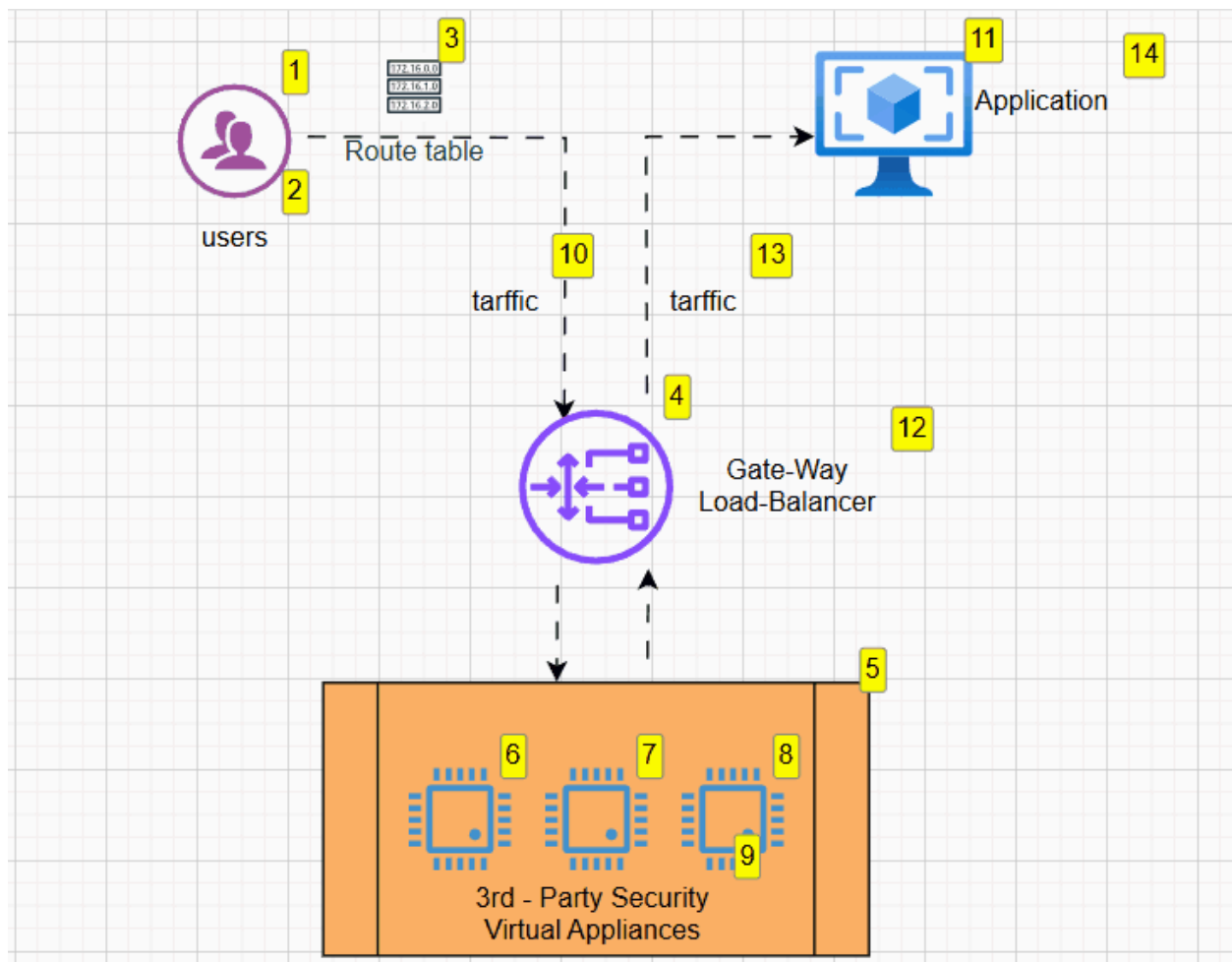


AWS Gateway Load Balancer (GWLB) – Explained

The **AWS Gateway Load Balancer (GWLB)** is a **Layer 3 (Network Layer)** load balancer designed **specifically to deploy, scale, and manage third-party virtual appliances** such as firewalls, intrusion detection and prevention systems (IDS/IPS), deep packet inspection systems (DPI), and traffic monitoring tools.

It combines **transparent traffic forwarding** with **load balancing**, enabling security appliances to scale elastically and integrate seamlessly into your AWS network.



Key Components of AWS Gateway Load Balancer

Component	Description
Gateway Load Balancer (GWLB)	Distributes traffic across virtual appliances using GENEVE encapsulation
Gateway Load Balancer Endpoint (GWLBe)	A VPC endpoint for privately forwarding traffic to the GWLB
Target Group	Group of virtual appliances (e.g., firewalls) the GWLB balances traffic to
GENEVE Protocol	Used for encapsulating traffic transparently; allows metadata tagging
Elastic Network Interface (ENI)	Interface used by virtual appliances to receive traffic from GWLB

How AWS Gateway Load Balancer Works

1. Traffic Flow Initiation

- A client request or packet destined for a resource passes through a **Gateway Load Balancer Endpoint (GWLBe)**.

2. Encapsulation

- The traffic is encapsulated using the **GENEVE protocol** by the GWLBe and forwarded to the GWLB.

3. Load Balancing

- GWLB distributes the traffic to one of the healthy **virtual appliances** in its **target group**.

4. Appliance Processing

- The virtual appliance (e.g., a firewall) inspects, modifies, or logs the traffic.

5. Return Path

- The appliance sends the processed packet back to the GWLB, which **decapsulates** and forwards it to its destination.

Use Cases

- Deploying third-party **next-gen firewalls** (e.g., Palo Alto, Fortinet, Check Point)
- Centralizing **network security inspection**
- **Intrusion detection/prevention**
- **Traffic mirroring and analysis**
- Enabling **service chaining** of security tools

Architecture Overview (Simplified)

Internet / VPC Traffic

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GWLB (Gateway Load Balancer Endpoint)

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GWLB (Encapsulates, Balances)

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Target Group (Security Appliances)

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Appliance processes and returns traffic

Benefits of GWLB

- **Transparent Deployment:** No need to change routing or application code.
- **Elastic Scaling:** Automatically scales security appliances.
- **High Availability:** Built-in failover and multi-AZ support.
- **Centralized Security:** Single point of traffic inspection.

 Comparison with Other Load Balancers

Feature	ALB	NLB	GWLB
OSI Layer	Layer 7 (App)	Layer 4 (Transport)	Layer 3 (Network)
Protocols Supported	HTTP, HTTPS	TCP, UDP, TLS	All IP traffic (L3)
Target Types	Web servers	EC2, IPs	Virtual appliances
Traffic Inspection	No	No	Yes (via appliances)
Use Case	Web apps	Low-latency apps	Security/middleware
